ROYAL UNIVERSITY OF BHUTAN

College of Science and Technology

Computing Technologies Department

Bachelor of Engineering in Software Engineering

IDE201 SOFTWARE ENGINEERING CAPSTONE PROJECT

PROJECT PROPOSAL

[CST Alumni Portal]

Group Number	Group 1
Academic Session	AS 2026
Academic Supervisor	Mr. Tsheten Dorji, Lecturer
External / Industry Supervisor	*[Name, Organisation — or N/A]*
Date of Submission	17 / 09 / 2026

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Declaration and Endorsement

We hereby declare that this proposal is our own original work, that all sources used have been duly acknowledged and cited, and that no part of this document has been submitted for assessment in any other module.

Group Leader	**Academic Supervisor**
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External Supervisor / Organisation Confirmation

> *Required only where the project is hosted by, or developed for, an external organisation. Per the module descriptor, confirmation of the proposed implementation plan by the external supervisor is due in Week 6 alongside this document. Mark N/A if the project has no external partner.*

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Confirmation of implementation plan	Signature: _____ Date: _____ Seal:

Project Identification

> *A one-page summary record for departmental filing. Mark any field that does not apply as N/A rather than deleting the row.*

Reference Number	*[For office use]*
Project Title	CST Alumni Portal
Keywords (max 5)	Alumni Portal, Alumni Management, Mentorship, Internship, Web Application
Project Status	New
Project Duration	Week 7 to Week 17–19 — approx. 13 weeks
Expected Start Date	17 / 08 / 2026

Organisations Involved

> *List any industrial, academic or funding partner and state precisely what each contributes — data, hardware, mentorship, deployment site, funding. Mark N/A if none.*

#	Organisation	Type (Industry / Academic / Funding)	Role and Contribution
1	College of Science and Technology	Academic	Project supervision, academic evaluation and institutional requirements
2			

Abstract

Communication and professional engagement among students, alumni, staff and administration at the College of Science and Technology (CST) are currently limited, making it difficult to maintain alumni connections, share career opportunities, provide mentorship and effectively manage alumni-related activities. Information may also be distributed across different communication channels, making it difficult for students and alumni to access relevant information and for the institution to coordinate alumni engagement.

This project proposes the development of a **CST Alumni Portal**, a centralized web-based platform that connects students and alumni while providing staff and administration with tools to manage alumni-related information and activities. The system will include a digital alumni yearbook, professional alumni profiles, mentorship and internship opportunities, messaging, application tracking, events, notifications and administrative management.

The system will be developed using **React.js and Tailwind CSS** for the frontend, **Node.js and Express.js** for the backend, and **MySQL with Prisma ORM** for database management.

The expected outcome is a functional and user-friendly portal that strengthens student–alumni engagement while enabling staff and administration to efficiently manage alumni information, opportunities and institutional activities through a centralized platform.

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1. Introduction

1.1 Background

Higher education institutions play an important role in preparing students for their transition into professional careers. This role extends beyond classroom learning, as students can benefit from professional networks, career guidance and connections with experienced graduates. Research on higher education mentoring indicates that alumni–student mentoring can provide benefits to both students and alumni and can contribute to stronger engagement with the university (Dollinger et al., 2019).

The **College of Science and Technology (CST)** is an engineering-focused institution under the Royal University of Bhutan that offers programmes in areas including civil engineering, electrical engineering, information technology, software engineering and architecture. CST also highlights practical learning, industry collaboration, internships and workshops as part of preparing students for professional careers. In this environment, maintaining meaningful connections among **students, alumni, staff and administration** can support the sharing of professional experiences, career guidance, opportunities and institutional activities. Research on graduate employability also identifies professional networks as an important element in the transition from higher education to work.

The proposed **CST Alumni Portal** aims to provide a centralized platform through which these stakeholders can interact and manage alumni-related activities. The proposed system will combine a digital alumni yearbook, professional alumni profiles, mentorship and internship opportunities, application tracking, events and notifications, while providing staff and administrators with appropriate management functions. The use of alumni mentoring is particularly relevant to the proposed system because research has found positive relationships between higher-education mentoring and areas of student career development, including career choice, networking and transition-related outcomes (Nabi et al., 2025).

1.2 Problem Statement

At the College of Science and Technology (CST), communication and information sharing among students, alumni, staff and administration are not sufficiently centralized, making it difficult to maintain continuous alumni engagement and access relevant professional information. Students may face difficulties connecting with experienced alumni for career guidance and finding relevant internship and job opportunities, while alumni have limited means to maintain professional connections and share their experiences and opportunities with current students. Staff and administration also face challenges in coordinating and managing alumni-related information and activities. As a result, relevant information may

remain scattered across different communication channels, leading to missed opportunities, inefficient communication and difficulty maintaining continuous engagement between students, alumni and the institution.

Desired State:

A centralized CST Alumni Portal providing:

- Role-based access for students, alumni, staff and administrators
- Alumni directory / digital yearbook with searchable profiles
- Alumni profile tracking — career history, research & publications, contributions
- Structured mentorship requests between students and alumni (no in-app messaging — contact happens via social/contact links on each profile)
- Centralized internship/opportunity postings with application tracking
- Event management with registration and notifications
- Admin dashboard for user, content and portal management

1.3 Stakeholders

The proposed CST Alumni Portal involves several stakeholder groups with different roles, interests and expectations. Identifying these stakeholders and their needs provides a basis for defining the system requirements and ensures that the proposed system supports both student–alumni interaction and institutional alumni management.

Stakeholder	Role / Interest	What they need from the system
1. Current Students	Primary users who seek career guidance, professional networking, mentorship and employment opportunities from the alumni community.	Access to alumni profiles and the digital yearbook, mentorship and internship opportunities, messaging services, application tracking, events and relevant notifications.
2. CST Alumni	Graduates who can contribute professional knowledge, experience, career guidance and employment opportunities to current students.	Ability to maintain professional profiles, connect with students, provide mentorship, communicate with students, share job and internship opportunities, and

		participate in alumni and career-related events.
3. CST Staff	Staff members who support and coordinate student–alumni engagement and institutional activities.	Administrative control over users, alumni information, opportunities, events, notifications and other system content, together with appropriate monitoring and reporting capabilities.
4. CST Administration	Institutional stakeholders responsible for overseeing alumni-related activities and the overall management of the portal.	Administrative control over users, alumni information, opportunities, events, notifications and other system content, together with appropriate monitoring and reporting capabilities.

Students and alumni are the **primary end users** of the proposed portal, while CST staff and the administration have important **coordination, management and institutional interests**. Their different requirements will be considered during system design to ensure that the portal supports meaningful student–alumni interaction while also providing appropriate institutional management capabilities.

1.4 Significance and Motivation

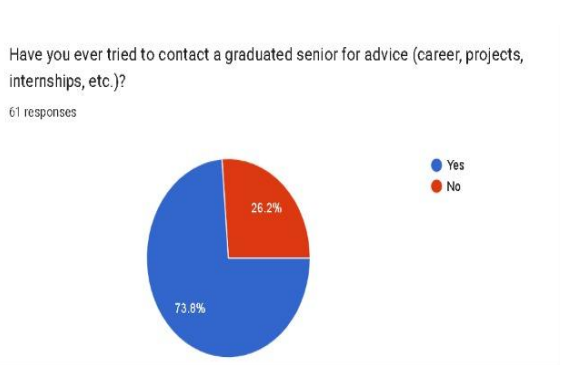
The proposed CST Alumni Portal is significant because it aims to strengthen engagement between **current students, alumni, staff and administration**. It will provide students with easier access to alumni networks, mentorship, career guidance and internships opportunities, while allowing alumni to remain connected with CST and contribute their knowledge and professional experience.

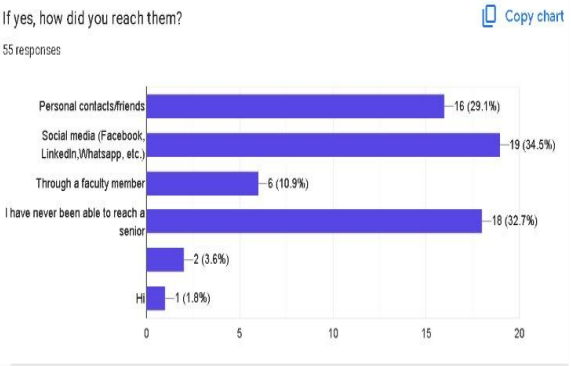
The project also has **societal and economic significance** by supporting knowledge sharing, professional networking and career development. A stronger connection between students and alumni can help students make informed career decisions and improve their awareness of professional opportunities.

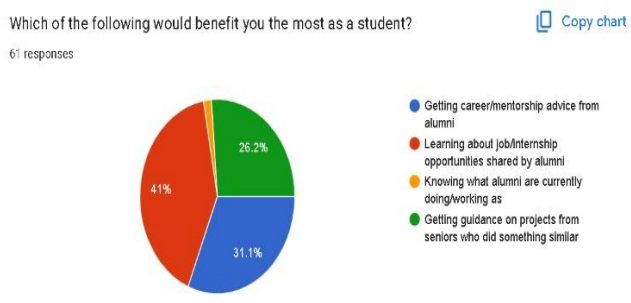
The need for the proposed system is supported by a survey conducted among **61 respondents**, which was used to understand the perceived need for a dedicated CST alumni platform and identify the areas that users consider important.

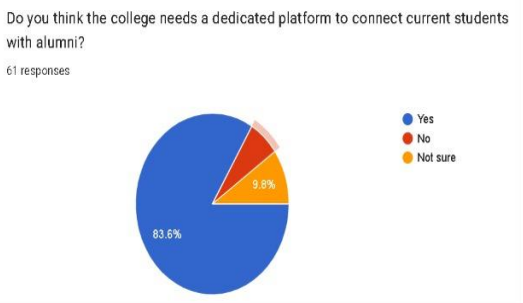
The project also provides sufficient scope for a **36-credit capstone project** by applying software engineering practices including requirements analysis, UI/UX design, database design, development, testing and deployment to address a real institutional need.

Evidence From the survey results









Findings	Result
Tried contacting alumni	73.8%
Feel interaction is insufficient	60.7%
Want a dedicated CST platform	83.6%
Value jobs/internships	41.0%
Value mentorship/career advice	31.1%

Figure 1 Evidence result from student survey

2. Literature Review and Related Work

2.1 Review of Existing Literature

Alumni engagement has become an important aspect of higher education because graduates can continue contributing to their institutions through **professional knowledge, mentoring, networking and institutional participation**. Mei et al. (2025), in a systematic review of 141 papers, identified alumni engagement as a multidimensional area closely associated with alumni relationships, student career development and personalized communication. Similarly, Dollinger et al. (2019), through their study of three university alumni mentoring programmes involving 197 participants, found that alumni mentoring can benefit both students and alumni while strengthening their engagement with the institution. These findings suggest that

effective alumni engagement requires opportunities for continued interaction rather than simply maintaining alumni records.

Mentoring and career development are particularly relevant to student–alumni relationships. Nuis et al. (2023) reviewed 106 studies on mentoring in higher education and found that mentoring can support student success, competence development and career development. Nabi et al. (2024), reviewing 73 studies published between 1986 and 2023, also found an overall positive relationship between higher-education mentoring and student career development, particularly in career choice and transition-related outcomes. However, both the literature and these reviews indicate that mentoring outcomes depend on the structure, context and method of delivery. This suggests that a student–alumni platform should provide a structured mechanism for connecting students with suitable alumni rather than relying only on general communication.

Professional networking is another important aspect of the transition from education to employment. De Schepper et al. (2023), in their systematic review of social networks during the transition from higher education to work, highlight the role of networks in providing access to employment-related information, resources and opportunities. Alumni networks can therefore provide students with access to professional experiences and connections that complement formal academic learning. A centralized platform can make these connections more accessible by allowing students to discover alumni based on their graduation year, programme and professional background.

The literature also indicates that **digital platforms can support alumni engagement by bringing different forms of interaction into a centralized environment**. However, simply providing an online alumni directory does not necessarily result in meaningful engagement. The effectiveness of such platforms depends on providing useful activities and appropriate structures for communication, mentoring and professional interaction. Therefore, the proposed CST Alumni Portal builds on these findings by bringing together **alumni information, professional networking, mentoring, communication and career opportunities** within a single platform.

Overall, the reviewed literature establishes that alumni engagement can contribute to **student career development, professional networking and stronger institutional relationships**, while structured mentoring can provide additional support during students' transition to employment. These findings provide the theoretical basis for developing a centralized student–alumni platform at CST.

#	Literature / Paper Reviewed	Key Finding	System Implication
1	Mei et al. (2025). Alumni engagement in higher education institutions: A comprehensive systematic literature review (SLR).	Alumni engagement benefits from personalized communication and long-term relationship management.	Alumni Profiles & Communication
2	Dollinger et al. (2019) University alumni mentoring programs: A win-win?	Alumni mentoring benefits both students and alumni.	Mentorship
3	Nuis et al. (2023) Conceptualizing mentoring in higher education: A systematic literature review.	Effective mentoring requires appropriate structure and design.	Mentorship Request Workflow
4	Nabi et al. (2024) The impact of mentoring in higher education on student career development: A systematic review and research agenda	Mentoring is positively associated with student career development.	Career Guidance
5	De Schepper et al. (2023) Social networks in the transition from higher education to work: A systematic review	Institutional networks can connect graduates to labour-market opportunities.	Networking & Jobs

2.2 Comparison of Existing Solutions

Existing Solution / Study	Technology Used	Strengths	Limitations
College Alumni Management System – Leki Yangdon et al. (2020)	CodeIgniter, PHP, MySQL	CST-specific alumni registration, management, communication, career information and events	Mainly focused on alumni management and college–alumni communication; limited student–alumni interaction and career workflow
JNEC Alumni Portal – Tshering & Wangchuk (2023)	Laravel, PHP, MariaDB	Alumni profiles, centralized information, search by graduation year/programme, events and networking	Limited documented mentorship and internship application workflow
Student and Alumni Web Portal (SAWP) – University of Zakho	PHP, MySQL, HTML, CSS, JavaScript, jQuery, AJAX	Connects students and alumni through a dedicated web application	Does not provide the full combination of mentorship, professional profiles and application tracking
Alumni Management System – Weeraketiya Rajapakshe Central College	PHP, MySQL	Registration, membership, events, news, donations and scholarships	Primarily focused on alumni association and administrative activities
Alumni Management System – College Crew	PHP, MySQL, HTML, CSS, JavaScript	Alumni registration, profiles, searching, events, announcements and discussion forum	Limited integration with structured student career-development activities
Sherubtse Alumni Association – Sherubtse College	College website / web-based platform	Alumni registration, networking, engagement and opportunities	Publicly available information does not show an integrated mentorship, application-tracking and messaging workflow

Existing Solution / Study	Technology Used	Strengths	Limitations
SCE Alumni Association – Samtse College of Education	College website / web-based system	Alumni registration, networking, professional development, events and recognition	Primarily association-based; limited publicly documented student–alumni career workflow

2.3 Research Gap

The reviewed literature and existing solutions demonstrate the value of alumni engagement, mentoring and professional networking, while existing alumni systems commonly provide registration, profiles, directories, events and institutional communication. However, the reviewed systems do not demonstrate a single integrated platform specifically designed to connect CST students, alumni, staff and administration while combining professional alumni profiles, a digital yearbook, structured mentorship, direct messaging, internship opportunities, application tracking, events and notifications. The previous CST alumni project provides a useful foundation, but the proposed CST Student Alumni Portal aims to extend this concept by placing greater emphasis on college–alumni interaction and career development. This gap directly supports the proposed objectives of developing a centralized platform for alumni networking, mentorship, career opportunities and institutional engagement.

3. Aim, Objectives and Scope

3.1 Aim

To develop a centralized web-based Alumni Portal for the College of Science and Technology (CST) that connects students, alumni, staff and administration through professional networking, mentorship and internship opportunities, communication and effective alumni engagement.

3.2 Objectives

- Provide role-based access to the portal for students, alumni, staff, and administrators, so each user group can log in and use features relevant to their role.
- Establish a secure alumni directory that allows students and staff to search and view alumni academic and professional information.
- Enable direct mentorship requests and internship postings between alumni and students, so students can find mentors and apply for opportunities within the platform.
- Support event management, notifications, and application tracking, so users stay informed and can monitor the status of their mentorship/internship applications.
- Validate the portal's usability and functionality through testing with target users, ensuring the system meets the needs identified in the survey and literature review.

3.3 Scope

In scope:

The proposed CST Student Alumni Portal will cover the following:

- **User and access management:** Registration, login, password recovery, and profile editing for four roles — Student, Alumni, Staff, and Administrator. Each role will have access only to the dashboard functions relevant to its responsibilities (e.g., students cannot access administrator verification tools).
- **Alumni directory and yearbook:** A searchable and filterable listing of alumni profiles by name, graduation year, and programme, together with a batch-wise yearbook browsing view. Both views will use the same alumni profile data rather than being developed as separate modules.
- **Mentorship request:** Students can browse alumni who have opted in as mentors and view relevant mentor information, including their **degree or programme pursued, area of expertise, and professional background**. Students can then submit a structured mentorship request containing a short description of their request and their **area of interest**. Alumni can accept or decline requests, and users can view the

request status. The mentorship feature is limited to this request-and-response workflow and does not include in-portal chat or messaging. Once a request is accepted, the portal may provide the parties' contact email addresses so that further communication can take place outside the system.

- **Internships:** Alumni or staff can post internship opportunities containing a title, description, deadline, and eligibility requirements. Students can browse available opportunities and submit applications through the portal. Students can track their application through four stages: **Applied, Under Review, Shortlisted, and Accepted/Not Selected**.
- **Events and notifications:** Staff and administrators can create and publish events such as career talks, reunions, and workshops. Alumni can view and register for events. The system will provide in-app notifications for new events, registration confirmations, event reminders, mentorship request updates, and internship application-status updates. These notifications will help users remain informed and encourage continued participation in relevant CST activities.
- **Administrative management:** Staff and administrators can verify new alumni accounts, while student and staff accounts must also undergo the appropriate verification process before they can access the system. Administrators can oversee verification activities, manage inappropriate or incorrect content, manage event and internship postings, and view basic usage reports such as the number of active mentorship requests and applications received for internship postings.
- **Database and system evaluation**:
- User, profile, mentorship, internship, application, event, and notification data will be stored in a relational database. Role-based access control will restrict users' ability to view and modify information according to their assigned roles.

Out of scope:

The following are outside the scope of the proposed project:

- **Mobile application:** A dedicated Android or iOS application will not be developed.
- **Online recruitment:** The portal will not conduct interviews, recruitment decisions or employment processing.
- **Payment services:** Online payment, donation and financial transaction features will not be included.
- **External institution integration:** Direct integration with external college, university, government or employment databases will not be implemented.
- **Advanced communication services:** Video calls, voice calls and other real-time communication services will not be included.

- **Automated career matching:** AI-based job recommendations, career prediction or automated mentor matching will not be developed.
- **In-portal messaging and advanced communication:** No direct text-based messaging, real-time chat, threaded conversations, voice calling, or video calling will be developed. After a mentorship request is accepted, users may use the contact email provided by the portal to communicate through external channels. This feature is excluded because developing and securing an in-portal communication system would add complexity beyond the project's available timeline and the team's current technical level.

3.4 Deliverables

- **Working CST Alumni Portal** – A functional web-based system implementing the approved features and user roles.
- **Source Code Repository** – Complete frontend, backend and database source code maintained in a version-controlled repository.
- **Database and System Design** – Final ERD, system architecture and other relevant design diagrams.
- **Testing Report** – Documentation of functional and usability testing, including identified issues and their results.
- **Deployed System** – A deployed version of the portal accessible for demonstration and evaluation.
- **Final Project Report** – Complete documentation covering the requirements, design, development, testing, results and conclusion of the project.

4. Methodology and System Design

4.1 Development Methodology

The project will follow a structured software development approach consisting of the following stages:

- Requirements Gathering – Collect system requirements from stakeholders and existing documentation.
- Requirements Analysis – Analyse and organize the identified requirements into functional and non-functional requirements.
- Software Design – Design the system interface, database, application program and overall system structure.
- Implementation – Develop the frontend, backend and database according to the approved design.
- Testing – Test individual features and the integrated system to identify and correct errors.
- Deployment – Deploy the completed system for use and evaluation.
- Report – Document the development process, results, testing and final system.

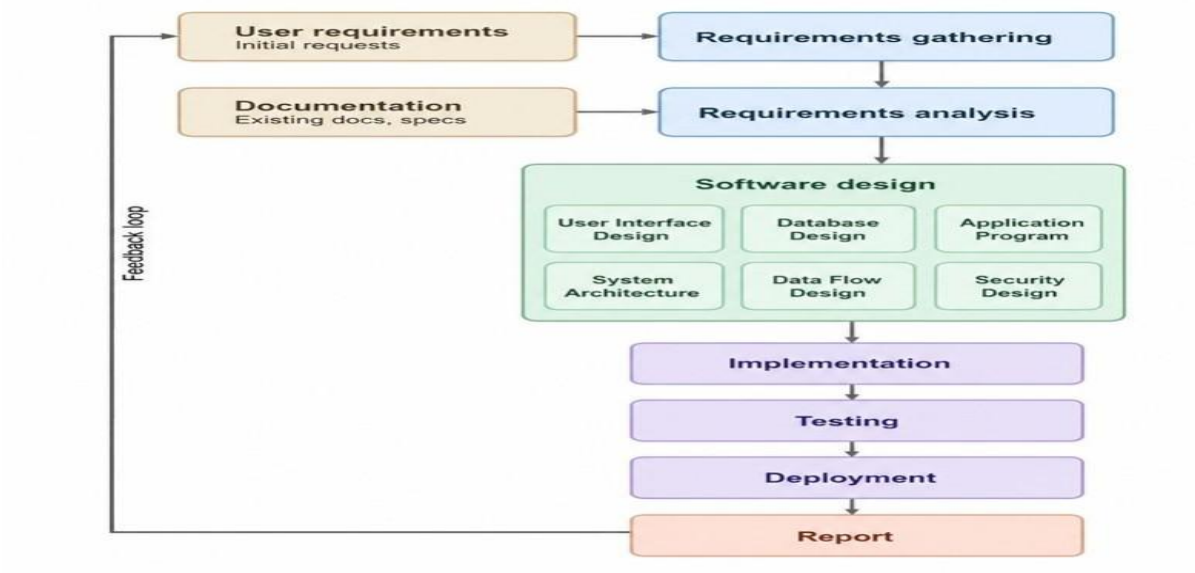


Figure 2 Methodology design

The development process will allow feedback to be considered at appropriate stages and improvements to be made before the final system is completed.

User workflow

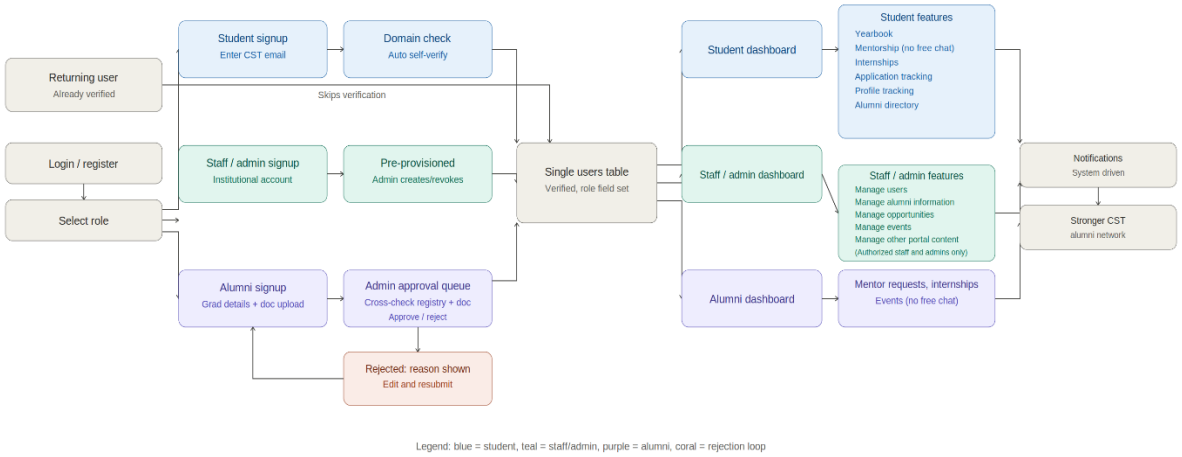


Figure 3 User Workflow

All users begin at a single login/register screen. Returning users who are already verified go straight to their dashboard. New users first select their role — Student, Staff/Admin, or Alumni — and each role follows a different path to prove who they are before getting access.

- Students sign up with their college email. The system checks that the email belongs to the school's domain and verifies them automatically, with no manual review needed.

- Staff and Admin accounts are not self-registered. A system admin creates these accounts directly, and can also revoke them later if someone leaves.

- Alumni go through the most careful check, since they no longer have an active college email. They submit their graduation details along with a supporting document. An admin reviews this against the alumni registry and either approves or rejects the application. If rejected, the applicant sees the reason and can edit and resubmit, rather than starting over from scratch.

Once verified, all users are stored in one single users table, with a role field marking whether they are a student, staff/admin, or alumni.

From there, each role reaches its own dashboard with matching features:

- Students get the yearbook, mentorship, internships, application tracking, profile tracking, and the alumni directory.

- Staff/Admin manage users, alumni information, opportunities, events, and other portal content.

- Alumni get mentorship requests, internships, and events.

Throughout the process, the system sends automatic notifications for updates and requests. The overall goal is a stronger, more connected CST alumni network, built on verified identity and clear role-based access.

4.2 System Design

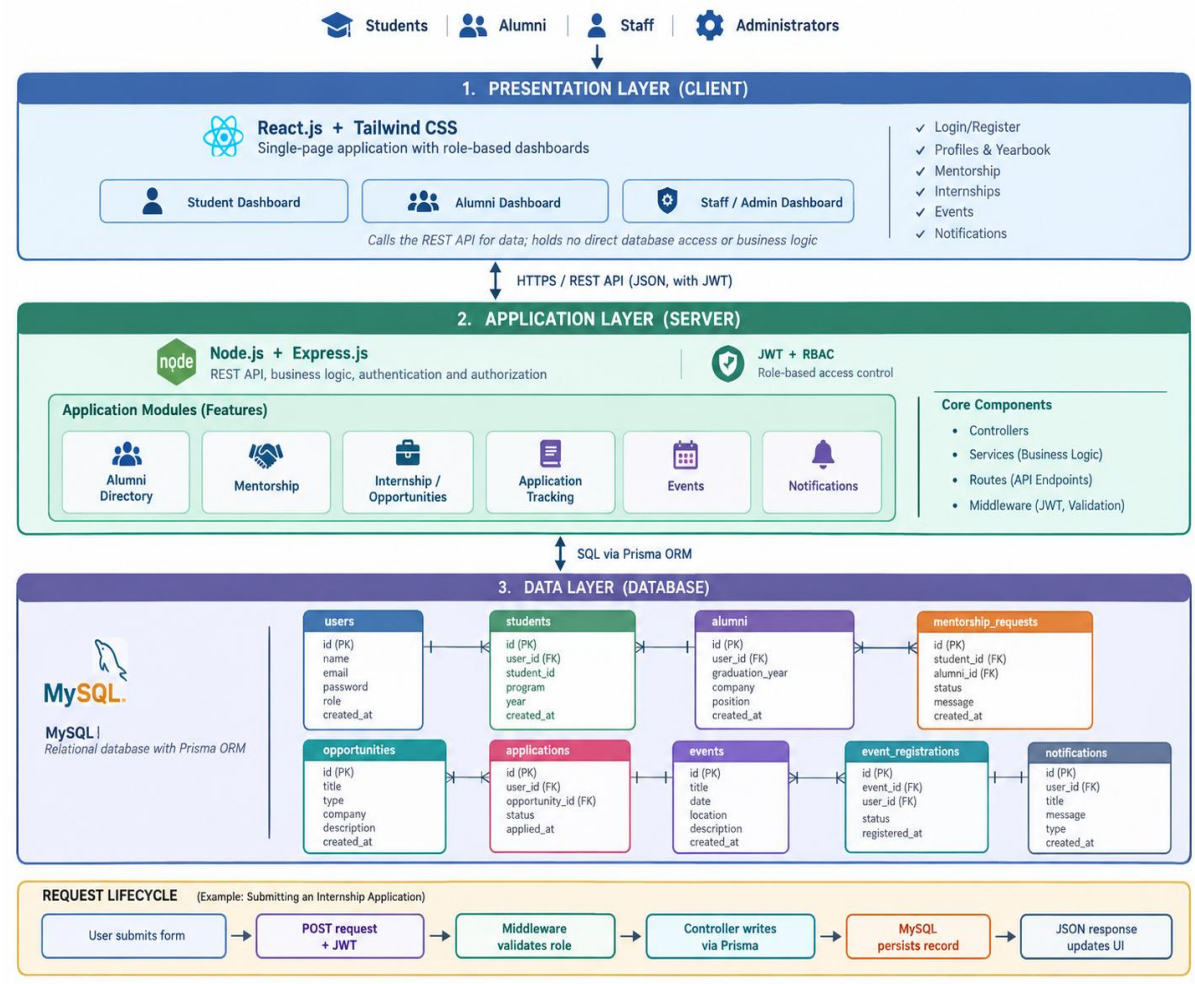


Figure 4 System Architecture

The system follows a three-tier client-server architecture, separating presentation, business logic, and data storage into independent layers.

The **presentation layer** is a React.js single-page application styled with Tailwind CSS. It renders role-based dashboards for students, alumni, staff and administrators, and communicates with the server exclusively through HTTPS calls to the REST API — it does not access the database directly and holds no business logic beyond input validation and UI state.

The **application layer**, built with Node.js and Express.js, owns all business logic. It is organized into modules — authentication and role-based access control (RBAC), alumni directory, mentorship, internship/opportunity management, application tracking, and events/notifications — each with its own routes and controllers. A JSON Web Token (JWT) is issued at login and attached to every subsequent request; middleware validates the caller's role before a request reaches a controller or the database, enforcing role-based access at a single, consistent point in the system.

A dedicated messaging module was considered but excluded from scope. Direct communication between students and alumni is expected to happen through existing social media and communication platforms, which alumni and students can already link from their own profile (e.g. LinkedIn, email, or other social handles) rather than the portal building and maintaining an in-house messaging system. This keeps the system's core focus on mentorship matching, opportunities and engagement, and avoids the added development, moderation and security overhead of running real-time communication in-house.

The **data layer** is a MySQL relational database, accessed only through Prisma ORM rather than raw SQL in application code, keeping queries type-safe and the schema as a single source of truth. The schema is normalized to third normal form (3NF) around core entities: User, Student, Alumni, MentorshipRequest, Opportunity, Application, Event, EventRegistration and Notification.

The bottom strip traces a concrete request end-to-end — a student submitting an internship application — showing how a single action moves through all three layers: the client submits the form, the request carries a JWT to the API, middleware checks the caller's role, the controller writes the record through Prisma, MySQL persists it, and a JSON response flows back to update the interface.

4.2.1 Database Design

A relational database will be designed using an Entity Relationship Diagram (ERD) to define the main entities, attributes, relationships and cardinalities of the system.

The database will be modelled as a normalized (3NF) relational schema, designed first as an Entity Relationship Diagram (Chen notation) and then implemented as a Prisma schema mapped 1:1 to MySQL tables. Core entities include:

- **User** (base identity, shared by all roles) specializing into **Student** and **Alumni**
- **Mentorship Request** (many-to-many between Students and Alumni, with a status field)
- **Opportunity** and **Application** (one Opportunity has many Applications; one User has many Applications)
- **Event** and **Event Registration** (many-to-many between Users and Events)

4.2.2 User Interface Design

The system interfaces will be designed using Figma before implementation. The designs will cover major functions such as login, dashboards, profiles, alumni directory/yearbook, mentorship, messaging, jobs and internships, applications, events and administration.

4.2.3 Testing and Evaluation

The completed system will be evaluated through:

- Functional testing – To verify that system features work as required.
- Integration testing – To verify communication between system components.
- Usability testing – To evaluate ease of use and navigation.
- Role-based access testing – To verify appropriate access for each user role.

5. Technical Implementation Plan

5.1 Technology Stack and Justification

Layer	Technology Selected	Justification / Alternatives Considered
Frontend	React.js, HTML, CSS, JavaScript	React.js is selected for its reusable component-based structure, which suits the portal's dashboards, profiles, forms and interactive modules. Vue.js was considered, but React aligns better with the team's existing knowledge and project requirements.
Backend	Node.js + Express.js	Node.js with Express.js is selected for developing RESTful APIs and handling authentication and system logic. Python was considered, but Node.js allows JavaScript to be used across both frontend and backend.
Database	MySQL	MySQL is selected because the portal contains strongly related data such as users, profiles, mentorships, messages, opportunities and applications. PostgreSQL was considered, but MySQL was chosen due to the team's existing database knowledge and experience.
Hardware / IoT	N/A	The project is a web-based system and does not require sensors, microcontrollers or other IoT hardware.
Hosting / Deployment	Render	Render is selected for deploying the web application and backend because it provides a simple cloud-based deployment approach suitable for a student project.
Testing Tools	Postman	Postman will be used to test REST API endpoints and verify backend functionality. It was selected because the team has prior experience using it for API testing.

5.2 Source Control and Code Review Workflow

- **Repository**: The project source code will be maintained in a private GitHub repository accessible to all group members and the project supervisor.
- **Branching Strategy:** A main branch will contain stable code. Each member will work on a separate **feature branch** and merge changes into main through pull requests.
- **Commit Convention:** Commits will use short and meaningful messages describing the change, e.g., Add alumni profile page or Fix login validation.
- **Pull Requests:** Members will create pull requests when their work is ready for integration. Each pull request must be reviewed by **at least one other group member** before merging.
- **Code Review:** Reviewers will check functionality, code quality, consistency and whether the changes meet the assigned requirements. The reviewer will request corrections where necessary before approval.
- **CI / Automation:** GitHub Actions will automatically run the project's **build and tests** on pull requests and changes pushed to the repository. Code will only be merged after the required checks pass.

Workflow:

Feature Branch → Commit → Pull Request → Code Review → CI Build/Test → Approval → Merge to `main`

5.3 Resource Requirements

> *Hardware, licences, cloud credits, devices. Per the module descriptor, requirements are verified by the supervisor and project coordinator before being forwarded for budget approval so itemise realistically, with indicative costs. Mark N/A if the project needs no procurement.*

Item / Specification	Quantity	Est. Unit Cost	Justification
Development laptops	5	N/A	Existing team devices will be used
Internet Access	5	N/A	Required for development and collaboration
Cloud hosting	1	N/A	Free/available hosting tier for project deployment
Software Licences	N/A	N/A	Open-source/free development tools will be used

6. Feasibility and Impact

6.1 Target Users and Beneficiaries

- **Current Students:** Use the portal to find alumni, profile tracking, request mentorship, access career guidance, discover internship opportunities and participate in events.
- **Alumni:** Maintain professional profiles, connect with students and other alumni, provide mentorship, and share career opportunities.
- **CST Administration:** Manage alumni engagement, events, announcements and institutional communication.
- **System Administrator:** Manage user accounts, verification, system security and portal maintenance.

Primary beneficiaries: Current students and alumni, with CST administration benefiting from centralized alumni information and communication.

6.2 Feasibility Analysis

Technical Feasibility

The system is technically feasible using the proposed technologies: React.js, Node.js/Express.js, MySQL and Prisma ORM. These technologies are widely available and suitable for developing a web-based portal. The project will use GitHub for source control and can be deployed using a cloud hosting service such as Render.

Operational Feasibility

The system is operationally feasible because it provides a centralized platform for activities such as alumni profiles, profile tracking, mentorship and internships, applications, events and notifications. Different user roles will have access to functions relevant to their responsibilities, allowing the system to support students, alumni and administrators.

Economic Feasibility

The project is economically feasible because the proposed development technologies and tools are primarily **free or open source**. Existing team computers will be used for development and testing, minimizing hardware costs. Initial deployment can use available cloud hosting options with minimal or no additional cost.

6.3 Scalability and Commercialisation Potential

Scalability

The proposed portal can initially serve CST students, alumni, staffs and administration and can later be expanded to support a larger number of users. The architecture can also be adapted for other colleges within the Royal University of Bhutan or similar educational institutions by configuring institution-specific users, profiles and administrative functions.

Commercialisation / Viability

The most realistic model is **institutional adoption**, where a college or university operates the portal for its students and alumni. Future sustainability could involve an institutional licensing or subscription model covering hosting, maintenance and technical support. However, commercialisation is outside the scope of the current student project and would require further validation with institutions.

Limitations

The initial system is designed for CST and will be developed as a pilot. Scalability to multiple institutions would require additional infrastructure, customization, maintenance and user-support resources.

6.4 Ethical, Legal and Data Considerations

The portal will handle personal and professional information such as **names, contact details, job history, contributions, publications, projects of alumni and employment information**, so user privacy and consent must be considered. Users should provide consent before their information is stored and displayed, and access to information should be controlled according to user roles. Passwords and sensitive information must be securely stored, while third-party libraries and software components will be used according to their respective licences. The interface should also consider accessibility and ease of use for different users.

7. Expected Outcomes

No.	Expected Outcome	Related Objective	Demonstration in Final Presentation
1	A functional, user-friendly web-based Alumni Portal with role-based access for students, alumni, staff and administrators.	Objective 1	Demonstrate login and the different functions available to each user role.
2	Secure user profiles and a searchable alumni directory containing academic and professional information.	Objective 2	Demonstrate profile creation, profile management and alumni directory search.
3	Integrated mentorship and internship opportunities to support student–alumni interaction and career development.	Objective 3	Demonstrate finding/requesting a mentor and viewing/applying for internship opportunities.
4	Event management, notifications and application tracking to support continuous engagement between students, alumni and CST.	Objective 4	Demonstrate event creation/registration, notifications and application status tracking.
5	A tested portal that meets the identified functional and usability requirements.	Objective 5	Present functional and usability testing results and demonstrate tested system features.

8. Project Plan

8.1 Milestones

#	Milestone / Deliverable	Target Week	Responsible Member(s)
M1	Requirements gathered, analysed and finalized. Project proposal approved	Week 5-6	Tandin Tashi Gyeltshen (Lead), All Members
M2	UI/UX, database and application design completed	Week 8	Choki Wangchuk, Pema Yangki Dorji, Khamsum Yoezer
M3	Core features implemented	Week 14	Khamsum Yoezer, Pema Yangki Dorji, Choki Wangchuk
M4	System integration and functional testing completed	Week 16	Pelden Choda Wangchuk, Khamsum Yoezer
M5	Final report and project documentation completed	Week 16	Tandin Tashi Gyeltshen (Lead), All Members
M6	Final prototype, testing and presentation completed	Week 17-19	All Members

8.2 Risk Summary

Risk Area	Low	Medium	High
Technical risk		☑	
Schedule / timing risk		☑	
Resource and budget risk	☑		
External dependency risk	☑		

Risk Justification

Technical Risk – Medium
The project requires integration of frontend, backend, database, authentication, mentorship, internships, applications and event management. Integration issues may require additional development and testing time.

Schedule / Timing Risk – Medium
The implementation, integration and testing activities are scheduled within a limited project period. Delays in one development area may affect subsequent activities.

Resource and Budget Risk – Low
The team will use existing computers and mainly free/open-source development tools. No major hardware or software procurement is expected.

External Dependency Risk – Low
The project has limited dependence on external systems. Internet access and cloud hosting may affect development or deployment, but they are not expected to create major project risks.

9. References

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